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Top performers and top journals: Persistent concentration in scientific publishing

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Electronic Supplementary Materials

Table ESM 1 and the corresponding Figure ESM 1 show the share of articles published in journals in the 90-99 journal percentile rank, broken down by gender, discipline, and time period. This measure can be interpreted as the probability of publications to appear in top journals and it is a direct indicator of access to the publishing elite (a class of top performers).

A striking feature of our results is strong disciplinary heterogeneity. The differences between disciplines are much greater (descriptively) than the differences between women and men within the same discipline. Technical and engineering disciplines (e.g., ENER, ENG, COMP) are characterized by relatively high shares of 90-99 percentile rank articles, while in areas such as MATH, MED, CHEM, and PHYS, the shares of publications in top journals remain systematically lower. The patterns are stable over time (over the five periods) and indicate persistent structural differences in publication patterns between scientists from different disciplines.

Against this backdrop, gender differences in the shares are moderate. In many large disciplines, the share of 90-99 percentile articles for men is slightly shifted upwards relative to the distribution observed for women. In the table, this effect is presented using mean values, but the non-parametric Mann-Whitney test does not refer to differences in means, but to the hypothesis of convergence of entire distributions (more precisely: the absence of a systematic advantage of one distribution over another).

However, the scale of these shifts is small and it most often corresponds to differences of a few percentage points. The frequent statistical significance of the Mann-Whitney test results primarily from the large sample sizes, which increase the power of the test and allow even small differences in the position of the distributions to be detected, rather than from the existence of clear, qualitatively large gender inequalities.

An analysis of the distributions (Figure ESM 1) further indicates that intra-gender variation significantly exceeds inter-gender variation. In most disciplines, the distributions are strongly skewed and dominated by observations close to zero, which means that a significant proportion of authors, regardless of gender, publish in top journals very rarely or never. At the same time, a small group of authors achieve very high values, creating long right tails in the distributions.

Taken together, these results suggest that gender differentiates access to prestigious journals in a moderate way that is strongly dependent on the disciplinary context, while the key source of inequality remains the author's position in the productivity hierarchy and structural differences between disciplines.

Table ESM 1. Percentage of articles in journals from the 90-99 Scopus percentile ranks, by gender, together with the Mann-Whitney test

Period	Discipline	Women	Men	% women	Mean share for women top performers	Mean share for men top performers
1992-1997	AGRI	581	757	43.4	0.146	0.144
1992-1997	BIO	1096	1066	50.7	0.156	0.166**
1992-1997	CHEM	1016	1751	36.7	0.089	0.090**
1992-1997	CHEMENG	36	131	21.6	0.205	0.170
1992-1997	COMP	26	128	16.9	0.221	0.427*
1992-1997	EARTH	180	511	26.0	0.095	0.104
1992-1997	ENER	6	28	17.6	0.556	0.536
1992-1997	ENG	99	712	12.2	0.351	0.398
1992-1997	ENVI	126	196	39.1	0.268	0.195
1992-1997	MATER	217	535	28.9	0.246	0.310*
1992-1997	MATH	159	697	18.6	0.088	0.108
1992-1997	MED	2525	3081	45.0	0.102	0.115***
1992-1997	NEURO	116	86	57.4	0.053	0.125
1992-1997	PHARM	155	150	50.8	0.086	0.041
1992-1997	PHYS	597	2470	19.5	0.119	0.146***
1998-2003	AGRI	1182	1326	47.1	0.136*	0.099
1998-2003	BIO	1975	1453	57.6	0.165	0.169**
1998-2003	CHEM	1898	2418	44.0	0.105	0.114***
1998-2003	CHEMENG	82	230	26.3	0.103	0.110
1998-2003	COMP	31	251	11.0	0.443	0.285
1998-2003	EARTH	400	892	31.0	0.076	0.088*
1998-2003	ENER	16	64	20.0	0.328	0.431
1998-2003	ENG	186	1156	13.9	0.283	0.305
1998-2003	ENVI	308	370	45.4	0.179	0.160
1998-2003	MATER	386	765	33.5	0.187	0.235**
1998-2003	MATH	276	931	22.9	0.081	0.075
1998-2003	MED	5021	5360	48.4	0.108	0.110*
1998-2003	NEURO	200	135	59.7	0.078	0.108
1998-2003	PHARM	196	135	59.2	0.102*	0.059
1998-2003	PHYS	822	3170	20.6	0.127	0.151***
2004-2009	AGRI	2242	2086	51.8	0.129	0.108
2004-2009	BIO	3060	1887	61.9	0.175	0.180***
2004-2009	CHEM	2820	2894	49.4	0.114	0.122***
2004-2009	CHEMENG	154	309	33.3	0.161	0.084
2004-2009	COMP	104	553	15.8	0.200	0.170
2004-2009	EARTH	568	1139	33.3	0.091	0.089
2004-2009	ENER	44	171	20.5	0.173	0.204
2004-2009	ENG	437	2614	14.3	0.154	0.171
2004-2009	ENVI	645	660	49.4	0.155	0.129

2004-2009	MATER	740	1365	35.2	0.172	0.189*
2004-2009	MATH	429	1204	26.3	0.062	0.082**
2004-2009	MED	9722	8049	54.7	0.093	0.102***
2004-2009	NEURO	268	151	64.0	0.105	0.102
2004-2009	PHARM	225	136	62.3	0.127	0.103
2004-2009	PHYS	1051	3676	22.2	0.138	0.162***
2010-2015	AGRI	3944	3214	55.1	0.114	0.116*
2010-2015	BIO	4490	2528	64.0	0.175	0.190***
2010-2015	CHEM	3947	3506	53.0	0.115	0.123***
2010-2015	CHEMENG	166	307	35.1	0.112	0.133*
2010-2015	COMP	210	1203	14.9	0.168	0.162
2010-2015	EARTH	889	1621	35.4	0.107	0.116*
2010-2015	ENER	133	404	24.8	0.156	0.183
2010-2015	ENG	809	4615	14.9	0.159	0.129
2010-2015	ENVI	1352	1113	54.8	0.144	0.132
2010-2015	MATER	1387	2237	38.3	0.142	0.120
2010-2015	MATH	551	1584	25.8	0.080	0.081*
2010-2015	MED	14,736	10,638	58.1	0.091	0.101***
2010-2015	NEURO	386	233	62.4	0.085	0.121
2010-2015	PHARM	311	159	66.2	0.124	0.163
2010-2015	PHYS	1234	4387	22.0	0.139	0.136
2016-2021	AGRI	5287	3936	57.3	0.214	0.201
2016-2021	BIO	5846	2989	66.2	0.289	0.289*
2016-2021	CHEM	4497	3660	55.1	0.173***	0.173
2016-2021	CHEMENG	203	348	36.8	0.148	0.176
2016-2021	COMP	301	1456	17.1	0.348*	0.288
2016-2021	EARTH	1259	2023	38.4	0.225	0.207
2016-2021	ENER	325	704	31.6	0.459	0.443
2016-2021	ENG	1516	6272	19.5	0.260	0.260
2016-2021	ENVI	2144	1687	56.0	0.279	0.281
2016-2021	MATER	2149	2814	43.3	0.176	0.154
2016-2021	MATH	587	1621	26.6	0.137	0.132
2016-2021	MED	19,972	12,700	61.1	0.152	0.153***
2016-2021	NEURO	532	339	61.1	0.189	0.196
2016-2021	PHARM	370	155	70.5	0.137	0.150
2016-2021	PHYS	1429	4596	23.7	0.175	0.188**

Figure ESM 1. Percentage of articles in journals in the 90-99 Scopus journal percentile ranks by gender